

CSE 311:

Data Communication

Instructor:

Dr. Md. Monirul Islam

Signals and systems

Topics

- Definition, classification and properties of signals
- Examples of some useful signals
- Fourier Series of Periodic Signals

Properties of Fourier Transform

Time shifting property $g(t - t_0) \Leftrightarrow G(f) \exp(-j2\pi f t_0)$

Properties of Fourier Transform

Time shifting property $g(t - t_0) \Leftrightarrow G(f) \exp(-j2\pi f t_0)$

$$F[g(t - t_0)] = \int_{-\infty}^{\infty} g(t - t_0) e^{-j2\pi f t} dt$$

Properties of Fourier Transform

Time shifting property $g(t - t_0) \Leftrightarrow G(f) \exp(-j2\pi f t_0)$

$$\begin{aligned} F[g(t - t_0)] &= \int_{-\infty}^{\infty} g(t - t_0) e^{-j2\pi f t} dt \\ &= \int_{-\infty}^{\infty} g(\tau) e^{-j2\pi f (t_0 + \tau)} d\tau \quad \text{assuming } t = t_0 + \tau \end{aligned}$$

Properties of Fourier Transform

Time shifting property $g(t - t_0) \Leftrightarrow G(f) \exp(-j2\pi f t_0)$

$$\begin{aligned} F[g(t - t_0)] &= \int_{-\infty}^{\infty} g(t - t_0) e^{-j2\pi f t} dt \\ &= \int_{-\infty}^{\infty} g(\tau) e^{-j2\pi f (t_0 + \tau)} d\tau \quad \text{assuming } t = t_0 + \tau \\ &= e^{-j2\pi f t_0} \int_{-\infty}^{\infty} g(\tau) e^{-j2\pi f \tau} d\tau \end{aligned}$$

Properties of Fourier Transform

Time shifting property $g(t - t_0) \Leftrightarrow G(f) \exp(-j2\pi f t_0)$

$$\begin{aligned} F[g(t - t_0)] &= \int_{-\infty}^{\infty} g(t - t_0) e^{-j2\pi f t} dt \\ &= \int_{-\infty}^{\infty} g(\tau) e^{-j2\pi f (t_0 + \tau)} d\tau \quad \text{assuming } t = t_0 + \tau \\ &= e^{-j2\pi f t_0} \int_{-\infty}^{\infty} g(\tau) e^{-j2\pi f \tau} d\tau \\ &= e^{-j2\pi f t_0} G(f) \end{aligned}$$

Properties of Fourier Transform

Time shifting property $g(t - t_0) \Leftrightarrow G(f) \exp(-j2\pi f t_0)$

- Delaying the signal by t_0 seconds DOES NOT change the amplitude spectrum, but changes phase spectrum by $-2\pi f t_0$

Properties of Fourier Transform

Time shifting property $g(t - t_0) \Leftrightarrow G(f) \exp(-j2\pi f t_0)$

- Delaying the signal by t_0 seconds DOES NOT change the amplitude spectrum, but changes phase spectrum by $-2\pi f t_0$
- If $g(t)$ is synthesized by Fourier components (sinusoids of certain amplitudes and phases), $g(t-t_0)$ can be synthesized by same sinusoids delayed by t_0 .

$$\begin{aligned} g(t) &= \int_{-\infty}^{\infty} G(f) e^{j2\pi f t} df \\ &= \int_{-\infty}^{\infty} [G(f) \cos 2\pi f t + jG(f) \sin 2\pi f t] df \end{aligned}$$

Properties of Fourier Transform

Time shifting property $g(t - t_0) \Leftrightarrow G(f) \exp(-j2\pi f t_0)$

- Delaying the signal by t_0 seconds DOES NOT change the amplitude spectrum, but changes phase spectrum by $-2\pi f t_0$
- If $g(t)$ is synthesized by Fourier components (sinusoids of certain amplitudes and phases), $g(t - t_0)$ can be synthesized by same sinusoids delayed by t_0 .

Assume the cosine signal $\cos 2\pi f t$ delayed by t_0

$$\cos 2\pi f (t - t_0) = \cos(2\pi f t - 2\pi f t_0)$$

Properties of Fourier Transform

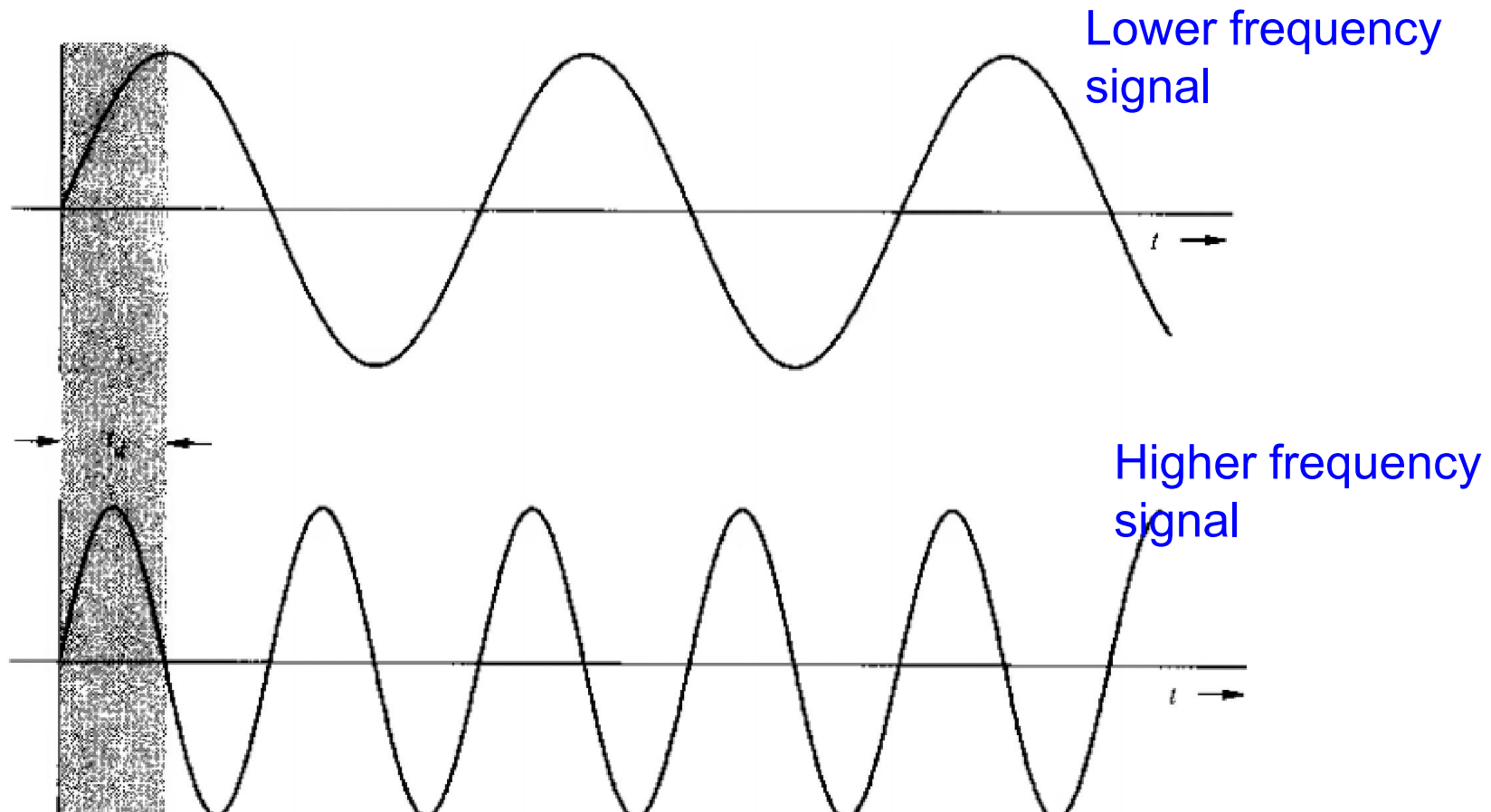
Time shifting property $g(t - t_0) \Leftrightarrow G(f) \exp(-j2\pi f t_0)$

$$\cos 2\pi f (t - t_0) = \cos(2\pi f t - 2\pi f t_0)$$

- Phase changes is linear function of f
- Higher frequency components undergoes proportionately higher phase shift

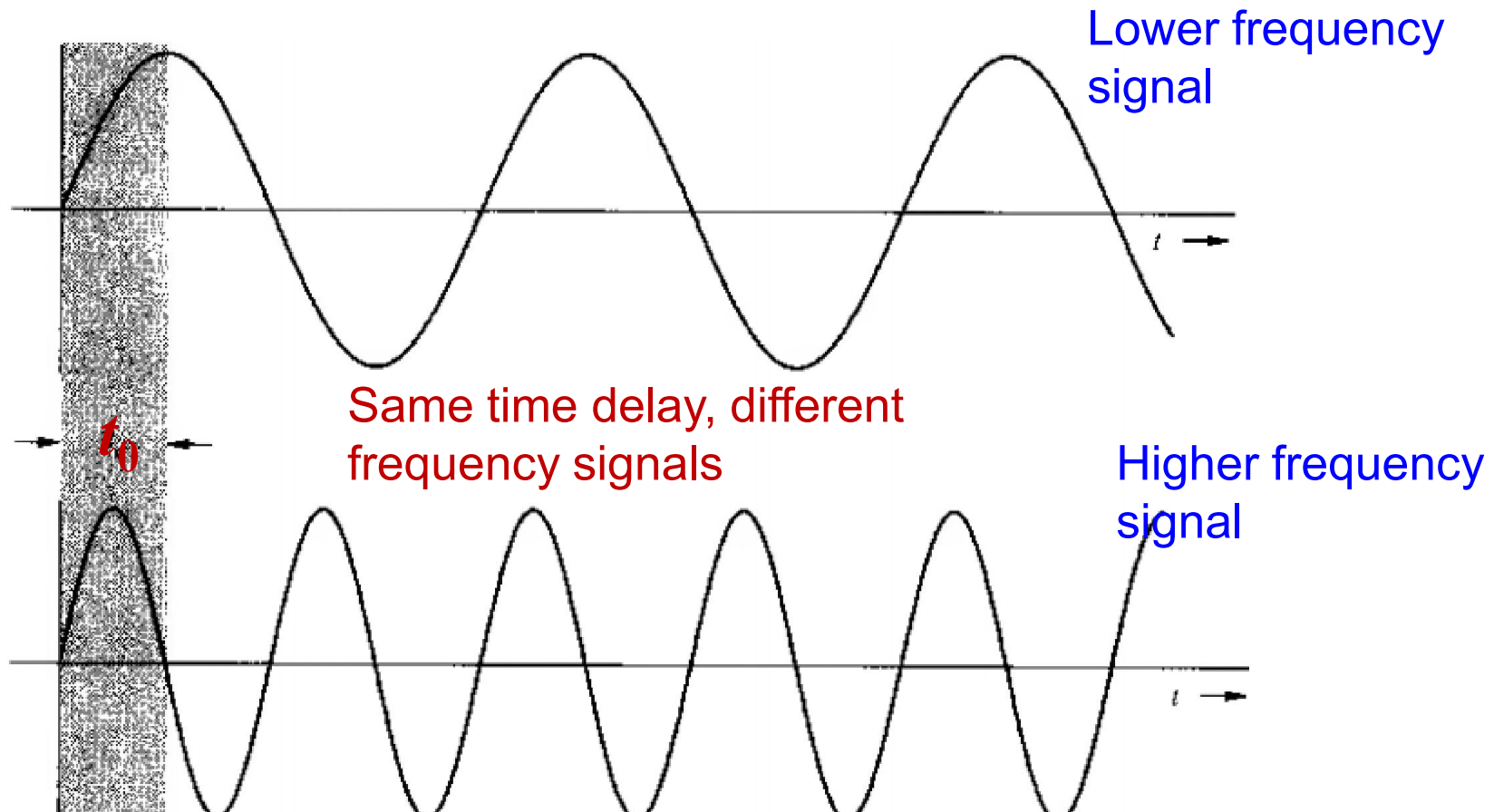
Properties of Fourier Transform

Time shifting property $g(t - t_0) \Leftrightarrow G(f) \exp(-j2\pi f t_0)$



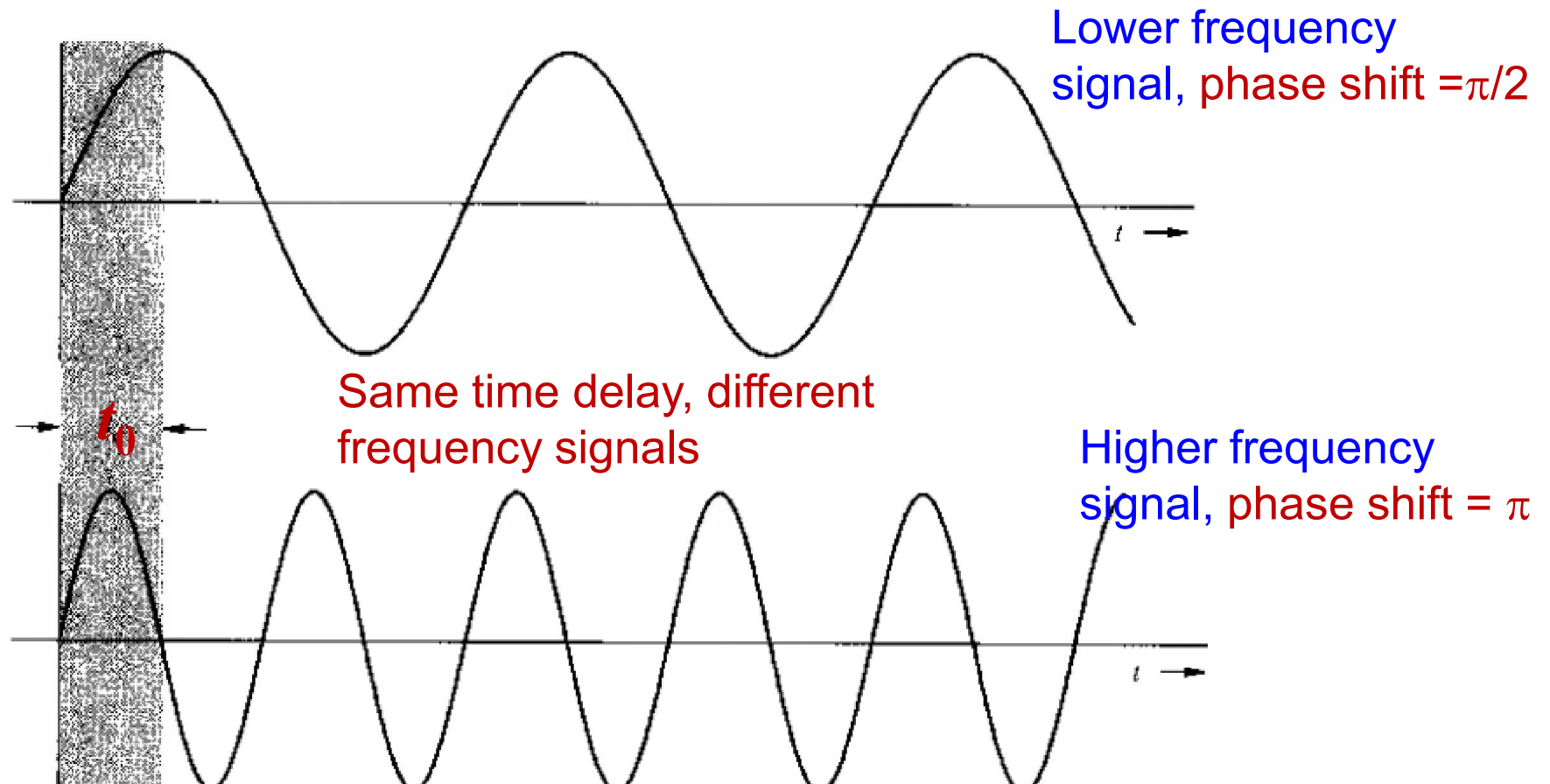
Properties of Fourier Transform

Time shifting property $g(t - t_0) \Leftrightarrow G(f) \exp(-j2\pi f t_0)$



Properties of Fourier Transform

Time shifting property $g(t - t_0) \Leftrightarrow G(f) \exp(-j2\pi f t_0)$



Properties of Fourier Transform

Frequency shifting
property

$$g(t) \exp(j2\pi f_0 t) \Leftrightarrow G(f - f_0)$$

Properties of Fourier Transform

Frequency shifting
property

$$g(t) \exp(j2\pi f_0 t) \Leftrightarrow G(f - f_0)$$

$$F[g(t)e^{j2\pi f_0 t}] = \int_{-\infty}^{\infty} g(t)e^{j2\pi f_0 t} e^{-j2\pi ft} dt$$

Properties of Fourier Transform

Frequency shifting
property

$$g(t) \exp(j2\pi f_0 t) \Leftrightarrow G(f - f_0)$$

$$\begin{aligned} F[g(t)e^{j2\pi f_0 t}] &= \int_{-\infty}^{\infty} g(t)e^{j2\pi f_0 t} e^{-j2\pi f t} dt \\ &= \int_{-\infty}^{\infty} g(t)e^{-j2\pi(f-f_0)t} dt \end{aligned}$$

Properties of Fourier Transform

Frequency shifting
property

$$g(t) \exp(j2\pi f_0 t) \Leftrightarrow G(f - f_0)$$

$$\begin{aligned} F[g(t)e^{j2\pi f_0 t}] &= \int_{-\infty}^{\infty} g(t)e^{j2\pi f_0 t} e^{-j2\pi f t} dt \\ &= \int_{-\infty}^{\infty} g(t)e^{-j2\pi(f-f_0)t} dt \\ &= G(f - f_0) \end{aligned}$$

Properties of Fourier Transform

Frequency shifting
property

$$g(t) \exp(j2\pi f_0 t) \Leftrightarrow G(f - f_0)$$

Replacing f_0 by $-f_0$

$$g(t) \exp(-j2\pi f_0 t) \Leftrightarrow G(f + f_0)$$

Properties of Fourier Transform

Frequency shifting
property

$$g(t) \exp(j2\pi f_0 t) \Leftrightarrow G(f - f_0)$$

Significance:

Multiplying $g(t)$ by $\exp(j2\pi f_0 t)$ shifts the spectrum by f_0 to a new location

Properties of Fourier Transform

Frequency shifting
property

$$g(t) \exp(j2\pi f_0 t) \Leftrightarrow G(f - f_0)$$

Significance:

Multiplying $g(t)$ by $\exp(j2\pi f_0 t)$ shifts the spectrum by f_0 to a new location

Recall FDM!

Properties of Fourier Transform

Frequency shifting
property

$$g(t) \exp(j2\pi f_0 t) \Leftrightarrow G(f - f_0)$$

Significance:

Multiplying $g(t)$ by $\exp(j2\pi f_0 t)$ shifts the spectrum by f_0 to a new location

Significance:

As $\exp(j2\pi f_0 t)$ is complex, $g(t)$ is multiplied by a real sinusoid $\cos(2\pi f_0 t)$ where,

$$\cos(2\pi f_0 t) = \frac{1}{2} (e^{j2\pi f_0 t} + e^{-j2\pi f_0 t})$$

Properties of Fourier Transform

Frequency shifting
property

$$g(t) \exp(j2\pi f_0 t) \Leftrightarrow G(f - f_0)$$

Therefore,
$$g(t) \cos(2\pi f_0 t) = \frac{1}{2} (g(t)e^{j2\pi f_0 t} + g(t)e^{-j2\pi f_0 t})$$

Properties of Fourier Transform

Frequency shifting
property

$$g(t) \exp(j2\pi f_0 t) \Leftrightarrow G(f - f_0)$$

Therefore, $g(t) \cos(2\pi f_0 t) = \frac{1}{2} (g(t) e^{j2\pi f_0 t} + g(t) e^{-j2\pi f_0 t})$

This results, $g(t) \cos(2\pi f_0 t) \Leftrightarrow \frac{1}{2} [G(f - f_0) + G(f + f_0)]$

Properties of Fourier Transform

Frequency shifting
property

$$g(t) \exp(j2\pi f_0 t) \Leftrightarrow G(f - f_0)$$

$$g(t) \cos(2\pi f_0 t) \Leftrightarrow \frac{1}{2} [G(f - f_0) + G(f + f_0)]$$

Significance:

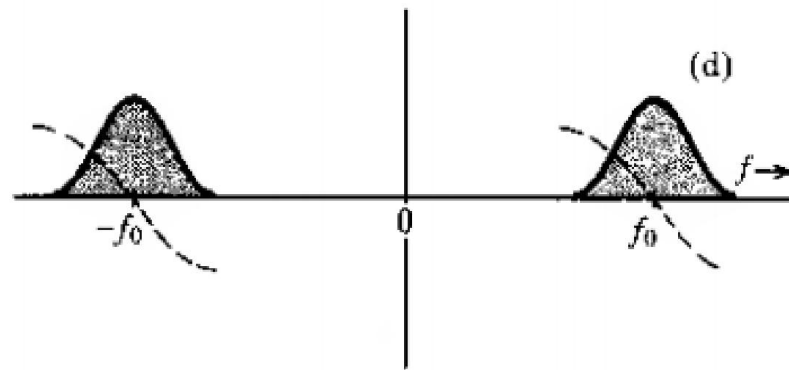
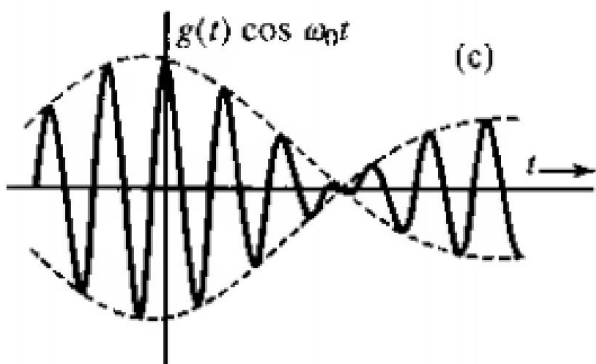
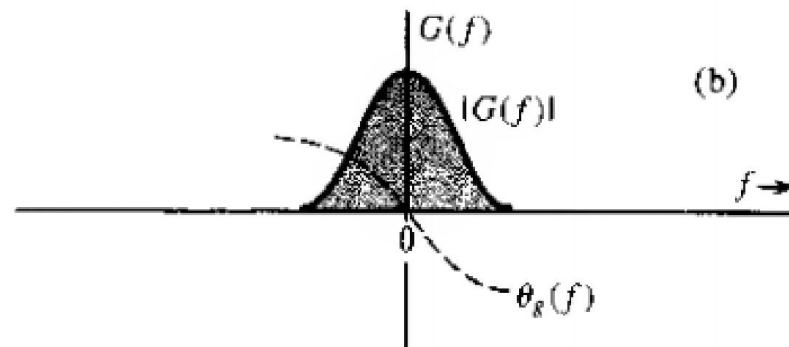
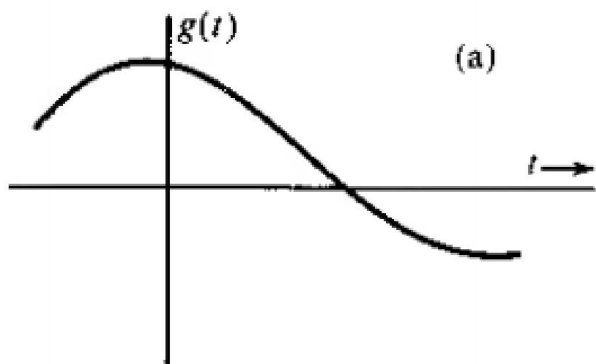
- Multiplying a sinusoid $\cos(2\pi f_0 t)$ by $g(t)$ means modulation of sinusoidal amplitude
- $\cos(2\pi f_0 t)$ is the carrier
- $g(t)$ is modulating signal
- This is amplitude modulation

Properties of Fourier Transform

Frequency shifting
property

$$g(t) \exp(j2\pi f_0 t) \Leftrightarrow G(f - f_0)$$

$$g(t) \cos(2\pi f_0 t) \Leftrightarrow \frac{1}{2} [G(f - f_0) + G(f + f_0)]$$



Properties of Fourier Transform

Frequency shifting
property

$$g(t) \exp(j2\pi f_0 t) \Leftrightarrow G(f - f_0)$$

$$g(t) \cos(2\pi f_0 t) \Leftrightarrow \frac{1}{2} [G(f - f_0) + G(f + f_0)]$$

Further phase change of modulated signal by θ_0 :

- Easily done by multiplying $\cos(2\pi f_0 t + \theta_0)$ instead of $\cos(2\pi f_0 t)$ by $g(t)$

Properties of Fourier Transform

Frequency shifting
property

$$g(t) \exp(j2\pi f_0 t) \Leftrightarrow G(f - f_0)$$

$$g(t) \cos(2\pi f_0 t) \Leftrightarrow \frac{1}{2} [G(f - f_0) + G(f + f_0)]$$

Further phase change of modulated signal by θ_0 :

- Easily done by multiplying $\cos(2\pi f_0 t + \theta_0)$ instead of $\cos(2\pi f_0 t)$ by $g(t)$

$$g(t) \cos(2\pi f_0 t + \theta_0) = \frac{1}{2} [g(t) e^{j2\pi f_0 t + j\theta_0} + g(t) e^{-j2\pi f_0 t - j\theta_0}]$$

$$= \frac{1}{2} [g(t) e^{j2\pi f_0 t} e^{j\theta_0} + g(t) e^{-j2\pi f_0 t} e^{-j\theta_0}]$$

Properties of Fourier Transform

Frequency shifting
property

$$g(t) \exp(j2\pi f_0 t) \Leftrightarrow G(f - f_0)$$

$$g(t) \cos(2\pi f_0 t) \Leftrightarrow \frac{1}{2} [G(f - f_0) + G(f + f_0)]$$

Further phase change of modulated signal by θ_0 :

- Easily done by multiplying $\cos(2\pi f_0 t + \theta_0)$ instead of $\cos(2\pi f_0 t)$ by $g(t)$

$$g(t) \cos(2\pi f_0 t + \theta_0) = \frac{1}{2} [g(t) e^{j2\pi f_0 t} e^{j\theta_0} + g(t) e^{-j2\pi f_0 t} e^{-j\theta_0}]$$

This means,

$$g(t) \cos(2\pi f_0 t + \theta_0) \Leftrightarrow \frac{1}{2} [G(f - f_0) e^{j\theta_0} + G(f + f_0) e^{-j\theta_0}]$$

Properties of Fourier Transform

Frequency shifting
property

$$g(t) \exp(j2\pi f_0 t) \Leftrightarrow G(f - f_0)$$

$$g(t) \cos(2\pi f_0 t + \theta_0) \Leftrightarrow \frac{1}{2} [G(f - f_0) e^{j\theta_0} + G(f + f_0) e^{-j\theta_0}]$$

$\theta_0 = 0$ means

$$g(t) \cos(2\pi f_0 t) \Leftrightarrow \frac{1}{2} [G(f - f_0) + G(f + f_0)]$$

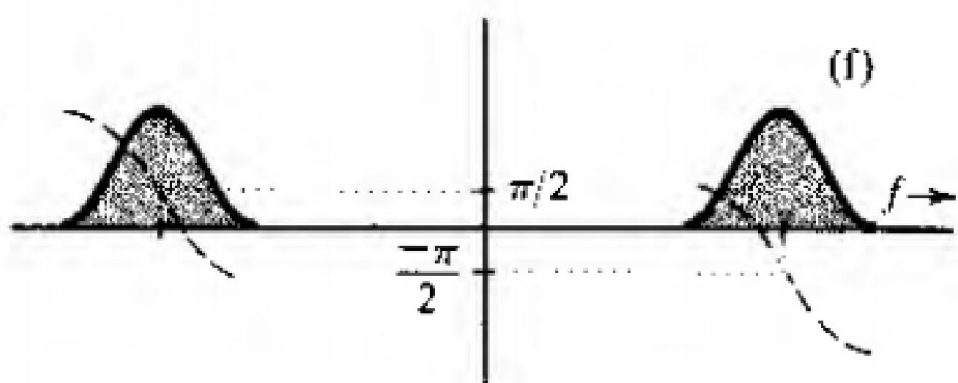
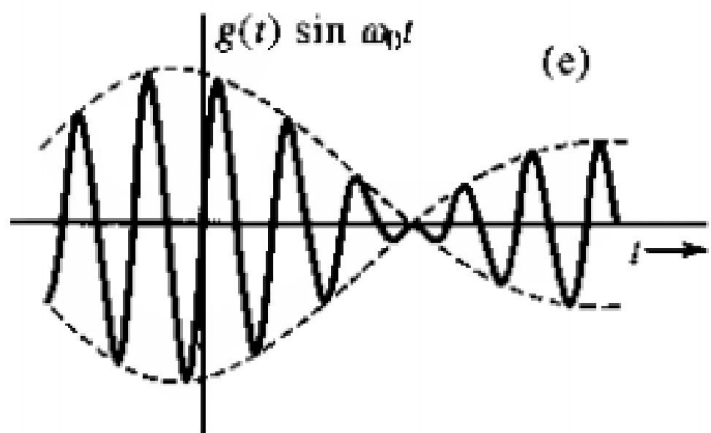
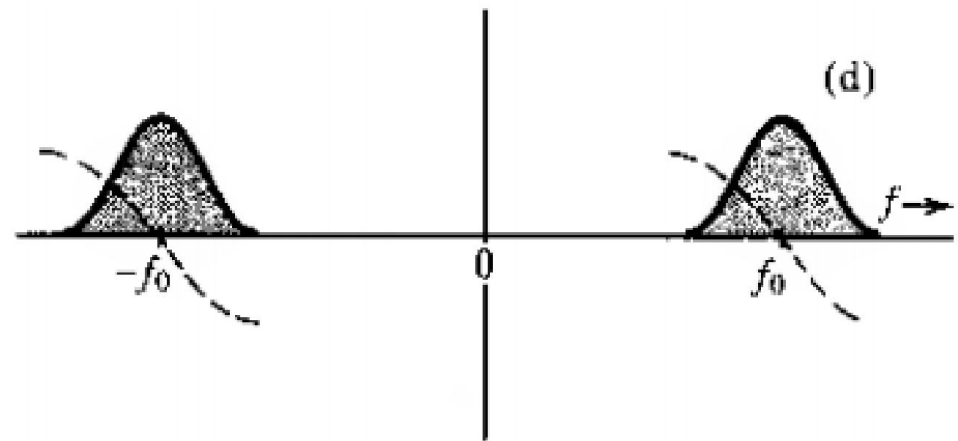
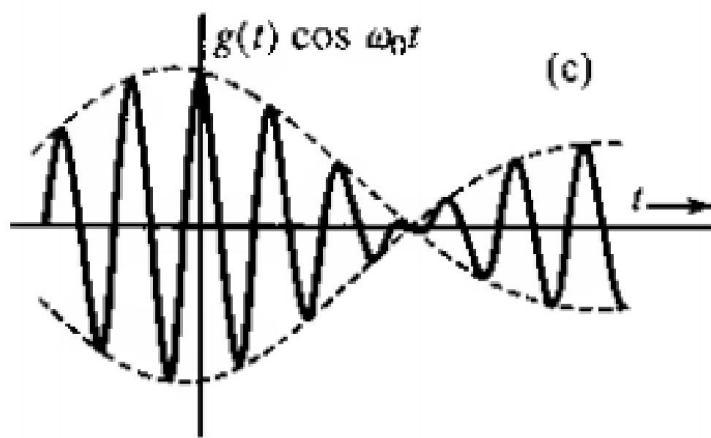
$\theta_0 = -\pi/2$ means

$$g(t) \cos(2\pi f_0 t - \frac{\pi}{2}) \Leftrightarrow \frac{1}{2} [G(f - f_0) e^{-j\frac{\pi}{2}} + G(f + f_0) e^{j\frac{\pi}{2}}]$$

Properties of Fourier Transform

Frequency shifting
property

$$g(t) \exp(j2\pi f_0 t) \Leftrightarrow G(f - f_0)$$



Properties of Fourier Transform

Frequency shifting
property in Modulation

$$g(t) \exp(j2\pi f_0 t) \Leftrightarrow G(f - f_0)$$

$$g(t) \cos(j2\pi f_0 t) \Leftrightarrow \frac{1}{2} [G(f - f_0) + G(f + f_0)]$$

Significance:

- A single common channel is used for multiple low band signals (e.g., multiple radio stations)
- Carriers of different frequencies are assigned to different transmission stations
- Each station shifts its signal to its own allocated band
- Simultaneous transmission is NO longer a problem

Properties of Fourier Transform

Convolution and Modulation Property

Time Convolution

$$g_1(t) * g_2(t) \Leftrightarrow G_1(f)G_2(f)$$

Modulation or frequency Convolution

$$g_1(t)g_2(t) \Leftrightarrow G_1(f) * G_2(f)$$

Properties of Fourier Transform

What is Convolution?

$$g_1(t) * g_2(t) = \int_{-\infty}^{\infty} g_1(\tau) g_2(t - \tau) d\tau$$

Properties of Fourier Transform

What is Convolution?

$$g_1(t) * g_2(t) = \int_{-\infty}^{\infty} g_1(\tau) g_2(t - \tau) d\tau$$

$$g_1(t) * g_2(t) = g_2(t) * g_1(t)$$

Properties of Fourier Transform

What is Convolution?

$$g_1(t) * g_2(t) = \int_{-\infty}^{\infty} g_1(\tau) g_2(t - \tau) d\tau$$

$$g_1(t) * g_2(t) = g_2(t) * g_1(t)$$

$$g_1(t) * g_2(t) = \int_{-\infty}^{\infty} g_1(\tau) g_2(t - \tau) d\tau = \int_{-\infty}^{\infty} g_2(\tau) g_1(t - \tau) d\tau$$

Properties of Fourier Transform

Time Convolution
Property

$$g_1(t) * g_2(t) \Leftrightarrow G_1(f)G_2(f)$$

$$\begin{aligned}\mathfrak{F}\{g_1(t) * g_2(t)\} &= \int_{-\infty}^{\infty} \left[\int_{-\infty}^{\infty} g_1(\tau) g_2(t - \tau) d\tau \right] e^{-j2\pi ft} dt \\ &= \int_{-\infty}^{\infty} g_1(\tau) \left[\int_{-\infty}^{\infty} g_2(t - \tau) e^{-j2\pi ft} dt \right] d\tau\end{aligned}$$

Properties of Fourier Transform

Time Convolution
Property

$$g_1(t) * g_2(t) \Leftrightarrow G_1(f)G_2(f)$$

$$\begin{aligned}\mathfrak{F}\{g_1(t) * g_2(t)\} &= \int_{-\infty}^{\infty} \left[\int_{-\infty}^{\infty} g_1(\tau) g_2(t - \tau) d\tau \right] e^{-j2\pi ft} dt \\ &= \int_{-\infty}^{\infty} g_1(\tau) \left[\int_{-\infty}^{\infty} g_2(t - \tau) e^{-j2\pi ft} dt \right] d\tau\end{aligned}$$

$$\int_{-\infty}^{\infty} g_2(t - \tau) e^{-j2\pi ft} dt \text{ is FT of } \tau \text{-time shifted } g_2(t), \text{ e.g., } G_2(f) e^{-j2\pi f\tau}$$

Properties of Fourier Transform

Time Convolution
Property

$$g_1(t) * g_2(t) \Leftrightarrow G_1(f)G_2(f)$$

$$\begin{aligned}\mathfrak{F}\{g_1(t) * g_2(t)\} &= \int_{-\infty}^{\infty} \left[\int_{-\infty}^{\infty} g_1(\tau) g_2(t - \tau) d\tau \right] e^{-j2\pi ft} dt \\ &= \int_{-\infty}^{\infty} g_1(\tau) \left[\int_{-\infty}^{\infty} g_2(t - \tau) e^{-j2\pi ft} dt \right] d\tau \\ &= \int_{-\infty}^{\infty} g_1(\tau) G_2(f) e^{-j2\pi f\tau} d\tau = G_2(f) \int_{-\infty}^{\infty} g_1(\tau) e^{-j2\pi f\tau} d\tau \\ &= G_2(f) \int_{-\infty}^{\infty} g_1(t) e^{-j2\pi ft} dt = G_2(f) G_1(f)\end{aligned}$$

Properties of Fourier Transform

Modulation or frequency
Convolution Property

$$g_1(t)g_2(t) \Leftrightarrow G_1(f) * G_2(f)$$

Properties of Fourier Transform

Modulation or frequency
Convolution Property $g_1(t)g_2(t) \Leftrightarrow G_1(f) * G_2(f)$

$$\mathfrak{F}^{-1}\{G_1(f) * G_2(f)\} = \int_{-\infty}^{\infty} \left[\int_{-\infty}^{\infty} G_1(\tau) G_2(f - \tau) d\tau \right] e^{j2\pi ft} df$$

Properties of Fourier Transform

Modulation or frequency
Convolution Property $g_1(t)g_2(t) \Leftrightarrow G_1(f) * G_2(f)$

$$\begin{aligned}\mathfrak{F}^{-1}\{G_1(f) * G_2(f)\} &= \int_{-\infty}^{\infty} \left[\int_{-\infty}^{\infty} G_1(\tau) G_2(f - \tau) d\tau \right] e^{j2\pi ft} df \\ &= \int_{-\infty}^{\infty} G_1(\tau) \left[\int_{-\infty}^{\infty} G_2(f - \tau) e^{j2\pi ft} df \right] d\tau\end{aligned}$$

Properties of Fourier Transform

Modulation or frequency
Convolution Property $g_1(t)g_2(t) \Leftrightarrow G_1(f) * G_2(f)$

$$\begin{aligned}\mathfrak{F}^{-1}\{G_1(f) * G_2(f)\} &= \int_{-\infty}^{\infty} \left[\int_{-\infty}^{\infty} G_1(\tau) G_2(f - \tau) d\tau \right] e^{j2\pi ft} df \\ &= \int_{-\infty}^{\infty} G_1(\tau) \left[\int_{-\infty}^{\infty} G_2(f - \tau) e^{j2\pi ft} df \right] d\tau \\ &= \int_{-\infty}^{\infty} G_1(\tau) g_2(t) e^{j2\pi \tau t} d\tau = g_2(t) \int_{-\infty}^{\infty} G_1(\tau) e^{j2\pi \tau t} d\tau\end{aligned}$$

Properties of Fourier Transform

Modulation or frequency
Convolution Property $g_1(t)g_2(t) \Leftrightarrow G_1(f) * G_2(f)$

$$\begin{aligned}\mathfrak{F}^{-1}\{G_1(f) * G_2(f)\} &= \int_{-\infty}^{\infty} \left[\int_{-\infty}^{\infty} G_1(\tau) G_2(f - \tau) d\tau \right] e^{j2\pi ft} df \\ &= \int_{-\infty}^{\infty} G_1(\tau) \left[\int_{-\infty}^{\infty} G_2(f - \tau) e^{j2\pi ft} df \right] d\tau \\ &= \int_{-\infty}^{\infty} G_1(\tau) g_2(t) e^{j2\pi \tau t} d\tau = g_2(t) \int_{-\infty}^{\infty} G_1(\tau) e^{j2\pi \tau t} d\tau \\ &= g_2(t) \int_{-\infty}^{\infty} G_1(f) e^{j2\pi ft} df = g_2(t) g_1(t)\end{aligned}$$

Properties of Fourier Transform

Time Integration
Property

$$\int_{-\infty}^t g(\tau) d\tau \Leftrightarrow \frac{G(f)}{j2\pi f} + \frac{1}{2} G(0) \delta(f)$$

Properties of Fourier Transform

Time Integration
Property

$$\int_{-\infty}^t g(\tau) d\tau \Leftrightarrow \frac{G(f)}{j2\pi f} + \frac{1}{2} G(0) \delta(f)$$

Recall

$$u(t) = \begin{cases} 1 & \text{if } t \geq 0 \\ 0 & \text{if } t < 0 \end{cases}$$

Properties of Fourier Transform

Time Integration
Property

$$\int_{-\infty}^t g(\tau) d\tau \Leftrightarrow \frac{G(f)}{j2\pi f} + \frac{1}{2} G(0) \delta(f)$$

Recall

$$u(t) = \begin{cases} 1 & \text{if } t \geq 0 \\ 0 & \text{if } t < 0 \end{cases}$$

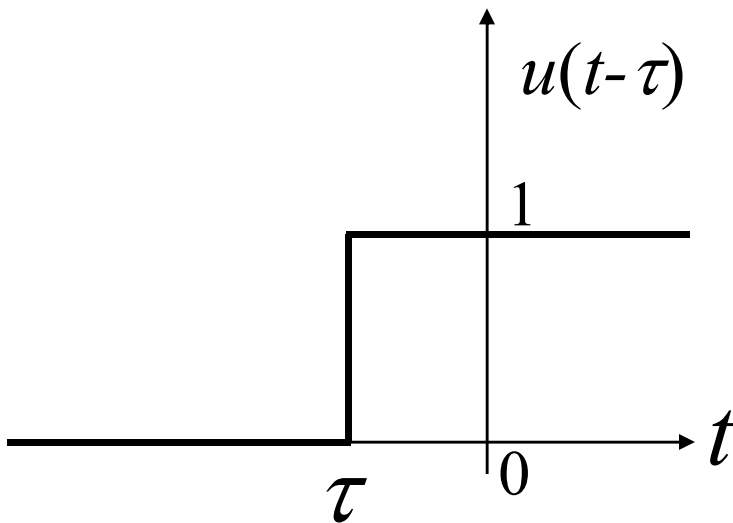
Therefore,

$$u(t - \tau) = \begin{cases} 1 & \text{if } t - \tau \geq 0 & \text{or } \tau \leq t \\ 0 & \text{if } t - \tau < 0 & \text{or } \tau > t \end{cases}$$

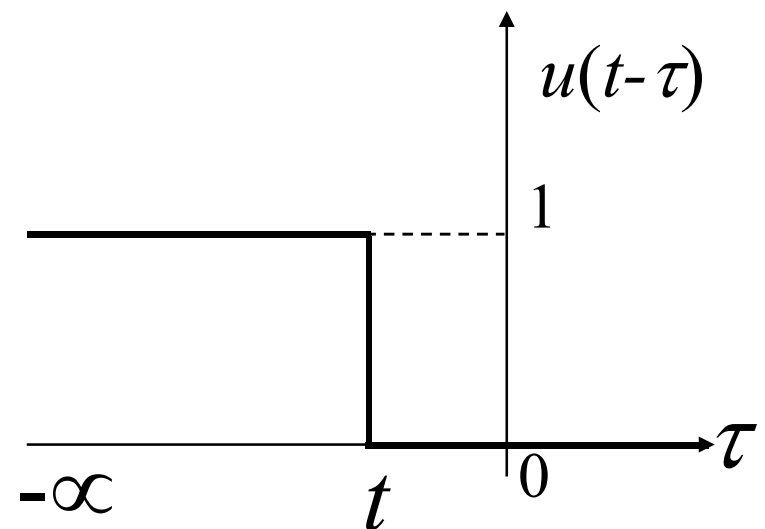
Properties of Fourier Transform

Time Integration
Property

$$\int_{-\infty}^t g(\tau) d\tau \Leftrightarrow \frac{G(f)}{j2\pi f} + \frac{1}{2} G(0) \delta(f)$$



$$u(t - \tau) = \begin{cases} 1 & \text{if } \tau \leq t \\ 0 & \text{if } \tau > t \end{cases}$$



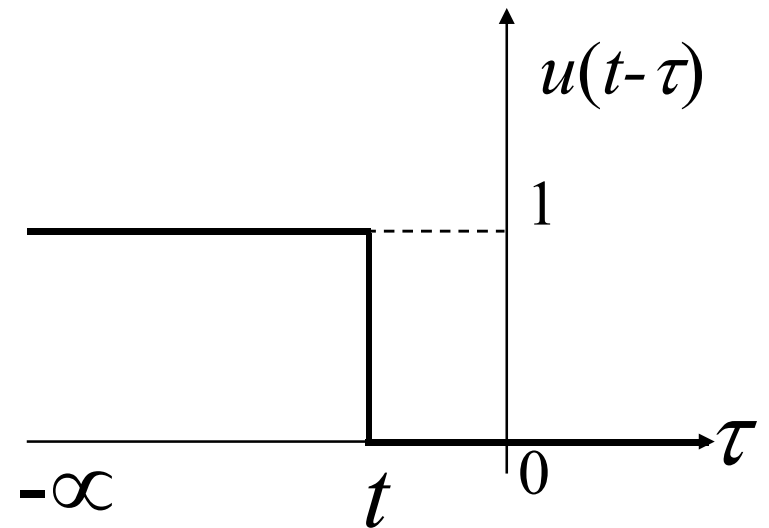
$$u(t - \tau) = \begin{cases} 1 & \text{if } t - \tau \geq 0 \\ 0 & \text{if } t - \tau < 0 \end{cases}$$

Properties of Fourier Transform

Time Integration
Property

$$\int_{-\infty}^t g(\tau) d\tau \Leftrightarrow \frac{G(f)}{j2\pi f} + \frac{1}{2} G(0) \delta(f)$$

$$g(t) * u(t) = \int_{-\infty}^{\infty} g(\tau) u(t - \tau) d\tau$$



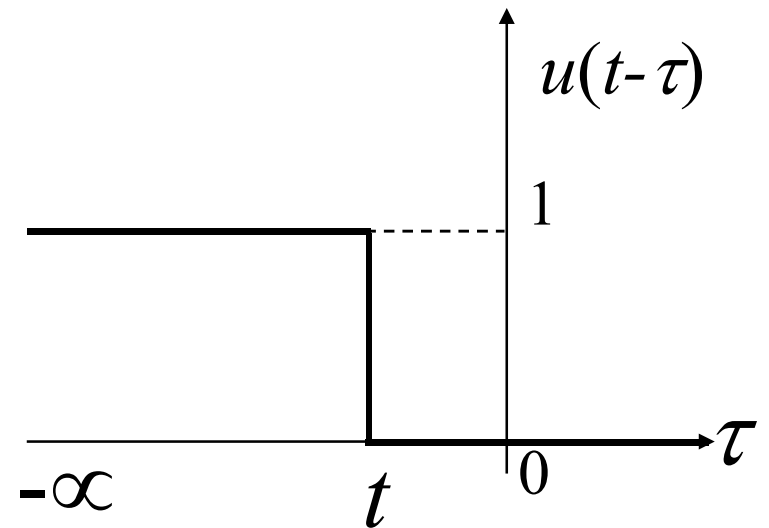
$$u(t - \tau) = \begin{cases} 1 & \text{if } t - \tau \geq 0 \\ 0 & \text{if } t - \tau < 0 \end{cases}$$

Properties of Fourier Transform

Time Integration
Property

$$\int_{-\infty}^t g(\tau) d\tau \Leftrightarrow \frac{G(f)}{j2\pi f} + \frac{1}{2} G(0) \delta(f)$$

$$g(t) * u(t) = \int_{-\infty}^{\infty} g(\tau) u(t - \tau) d\tau$$
$$= \int_{-\infty}^t g(\tau) u(t - \tau) d\tau + \int_t^{\infty} g(\tau) u(t - \tau) d\tau$$



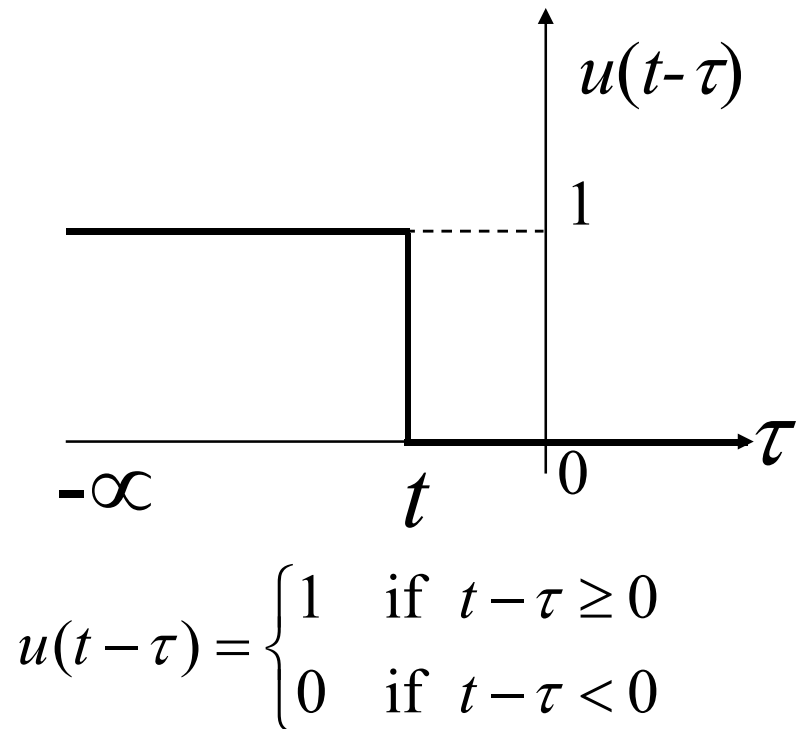
$$u(t - \tau) = \begin{cases} 1 & \text{if } t - \tau \geq 0 \\ 0 & \text{if } t - \tau < 0 \end{cases}$$

Properties of Fourier Transform

Time Integration
Property

$$\int_{-\infty}^t g(\tau) d\tau \Leftrightarrow \frac{G(f)}{j2\pi f} + \frac{1}{2} G(0) \delta(f)$$

$$\begin{aligned} g(t) * u(t) &= \int_{-\infty}^{\infty} g(\tau) u(t-\tau) d\tau \\ &= \int_{-\infty}^t g(\tau) u(t-\tau) d\tau + \int_t^{\infty} g(\tau) u(t-\tau) d\tau \\ &= \int_{-\infty}^t g(\tau) \cdot 1 d\tau + \int_t^{\infty} g(\tau) \cdot 0 d\tau = \int_{-\infty}^t g(\tau) d\tau \end{aligned}$$



Properties of Fourier Transform

Time Integration
Property

$$\int_{-\infty}^t g(\tau) d\tau \Leftrightarrow \frac{G(f)}{j2\pi f} + \frac{1}{2} G(0) \delta(f)$$

$$g(t) * u(t) = \int_{-\infty}^t g(\tau) d\tau$$

Properties of Fourier Transform

Time Integration
Property

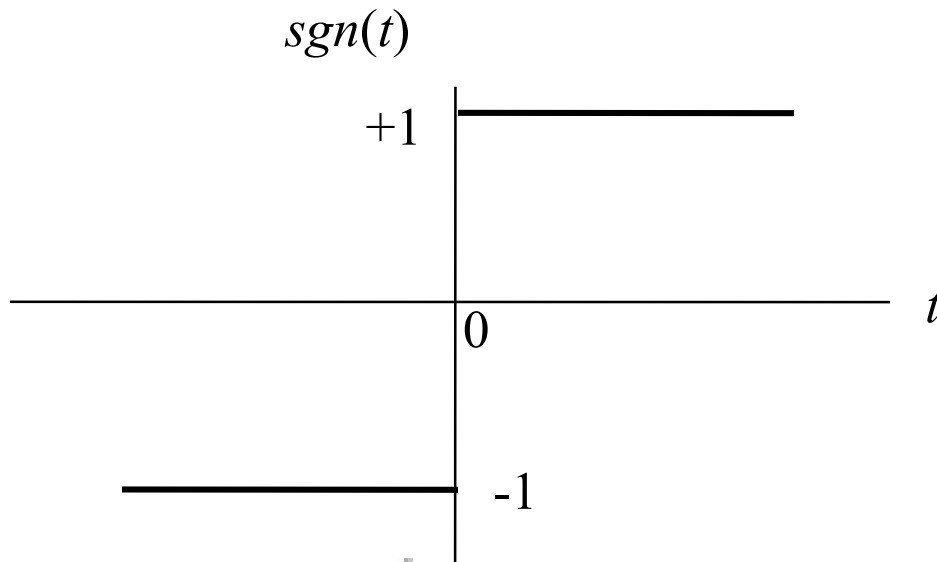
$$\int_{-\infty}^t g(\tau) d\tau \Leftrightarrow \frac{G(f)}{j2\pi f} + \frac{1}{2} G(0) \delta(f)$$

$$g(t) * u(t) = \int_{-\infty}^t g(\tau) d\tau$$

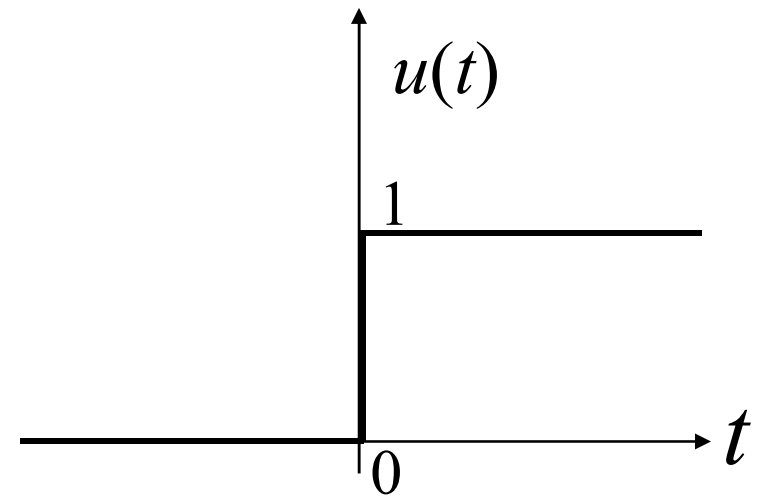
$$\int_{-\infty}^t g(\tau) d\tau = g(t) * u(t) \Leftrightarrow G(f)U(f)$$

Properties of Fourier Transform

$u(t)$ and $\text{sgn}(t)$ function



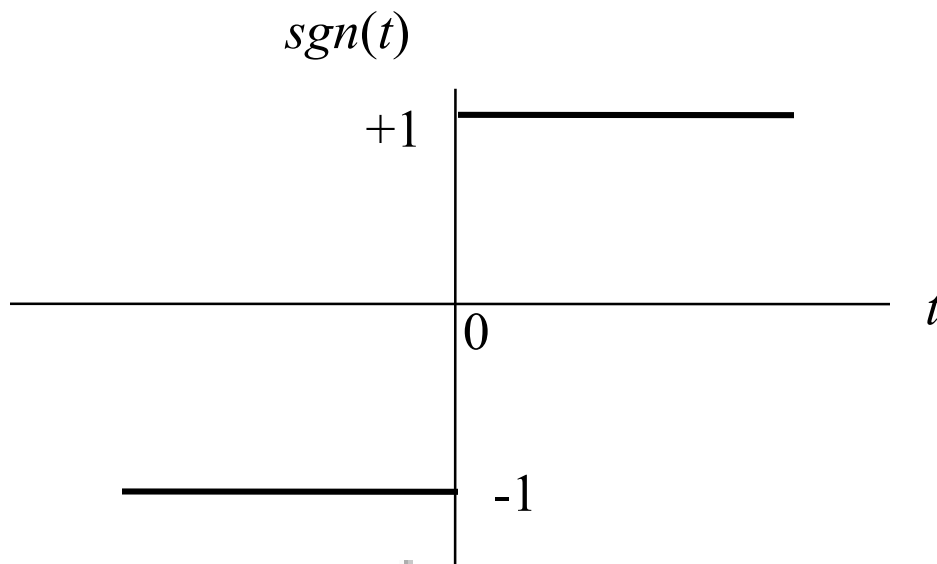
$$\text{sgn}(t) = \begin{cases} +1, & t > 0 \\ 0, & t = 0 \\ -1, & t < 0 \end{cases}$$



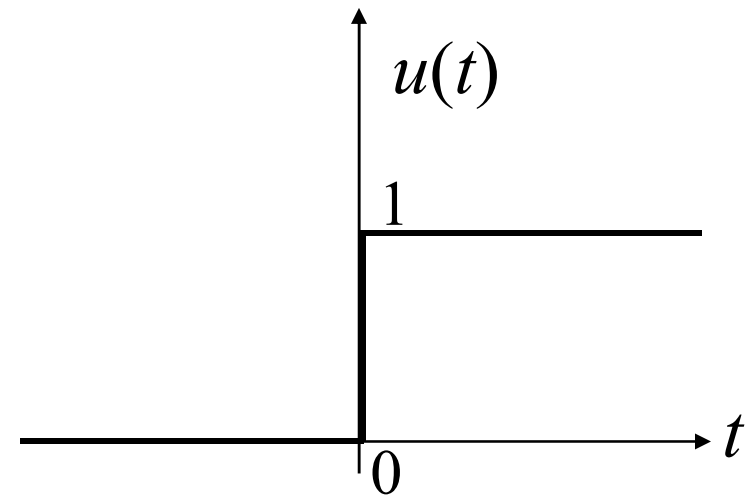
$$u(t) = \begin{cases} 1 & \text{if } t \geq 0 \\ 0 & \text{if } t < 0 \end{cases}$$

Properties of Fourier Transform

$u(t)$ and $\text{sgn}(t)$ function



$$\text{sgn}(t) = \begin{cases} +1, & t > 0 \\ 0, & t = 0 \\ -1, & t < 0 \end{cases}$$



$$u(t) = \begin{cases} 1 & \text{if } t \geq 0 \\ 0 & \text{if } t < 0 \end{cases}$$

Therefore, $u(t) = \frac{1}{2}(\text{sgn}(t) + 1)$

Properties of Fourier Transform

$u(t)$ and $\text{sgn}(t)$ function

$$u(t) = \frac{1}{2}(\text{sgn}(t) + 1)$$

Therefore,
$$U(f) = \frac{1}{2} \left(\frac{1}{j\pi f} + \delta(f) \right) = \frac{1}{j2\pi f} + \frac{1}{2} \delta(f)$$

Properties of Fourier Transform

Time Integration
Property

$$\int_{-\infty}^t g(\tau) d\tau \Leftrightarrow \frac{G(f)}{j2\pi f} + \frac{1}{2} G(0) \delta(f)$$

$$g(t) * u(t) = \int_{-\infty}^t g(\tau) d\tau$$

$$\int_{-\infty}^t g(\tau) d\tau = g(t) * u(t) \Leftrightarrow G(f)U(f)$$

$$U(f) = \frac{1}{2} \left(\frac{1}{j\pi f} + \delta(f) \right) = \frac{1}{j2\pi f} + \frac{1}{2} \delta(f)$$

Properties of Fourier Transform

Time Integration
Property

$$\int_{-\infty}^t g(\tau) d\tau \Leftrightarrow \frac{G(f)}{j2\pi f} + \frac{1}{2} G(0) \delta(f)$$

$$g(t) * u(t) = \int_{-\infty}^t g(\tau) d\tau$$

$$\int_{-\infty}^t g(\tau) d\tau = g(t) * u(t) \Leftrightarrow G(f)U(f)$$

$$= \frac{G(f)}{j2\pi f} + \frac{1}{2} G(f) \delta(f)$$

$$= \frac{G(f)}{j2\pi f} + \frac{1}{2} G(0) \delta(f)$$